TEXORA · SESSIONAL NOTES · PRACTICAL STUDY SHEET

Operational Technique of Knitting Machinery: Starting, Threading, Checking, Fault Finding and Safe Working

How a knitting machine is actually run: the pre-start inspection, correct threading, the start-up and stopping sequence, the needle and timing checks that keep it in tune, a fault-finding table for everything that goes wrong while running — and the safety rules that are never optional.

TE-3102 · KNITTING TECHNOLOGY SESSIONAL · STUDY 4 · LEVEL 3 TERM 1

1. Object

Object: to learn the operator’s craft — the order in which a competent knitter inspects, threads, starts, watches, stops and doffs a machine, the checks for needle condition and timing, and the systematic tracing of running faults. Technique is examined practically: you may be asked to thread a feeder, start and stop the machine, and diagnose a fault the instructor has set up. This sheet is the sequence to practise.

2. Before starting — the pre-start inspection

Never start a machine you have not walked around. Work through the list in this order; every item exists because it has broken needles or ruined fabric when skipped.

1. **Needles.** Run a fingertip (machine stopped) or a strip of waste fabric around the needle circle and feel for snags: a bent hook, burred stem or stiff latch catches at once. Replace a damaged needle with the same type and gauge, butt in the track, latch moving freely under its own spring — never oil a lint-stiff latch into “working”. On the flat machine check both beds and the latch-opening brushes; on the warp machine, sight along the bars for bent guides and damaged needles.
2. **Sinkers.** Bent or burred sinkers and lint packed in the sinker ring: a sinker that cannot move freely gives Study 2’s faults — fabric lifting at clearing, loops held back at knock-over.
3. **Cams.** Inch the machine by hand through one feeder and watch each needle rise fully, take the yarn and draw down evenly. A needle that jumps, hesitates or clicks has a damaged butt or a fouled cam

track.

4. **Lint and oil.** Clear lint from tricks, sinker ring and cam boxes (packed lint is the commonest cause of stiff needles). Check oil level, oil reaching the needle line as a faint film, and no drips where oil can reach fabric.
5. **Yarn.** Right fibre and count from the work ticket; cones sound on the creel, none with a damaged base or sloughed coils to snatch on unwinding.
6. **Fabric and take-down.** Previous fabric correctly on the needles or properly cleared; take-down rollers clean, nip even, path to the batching roll unobstructed, roll not over-full.
7. **Stop motions and guards.** Every stop motion in working position (never tied up or wedged — if you find one defeated, report it, never copy it), lamps working, guards in place, emergency stop free.

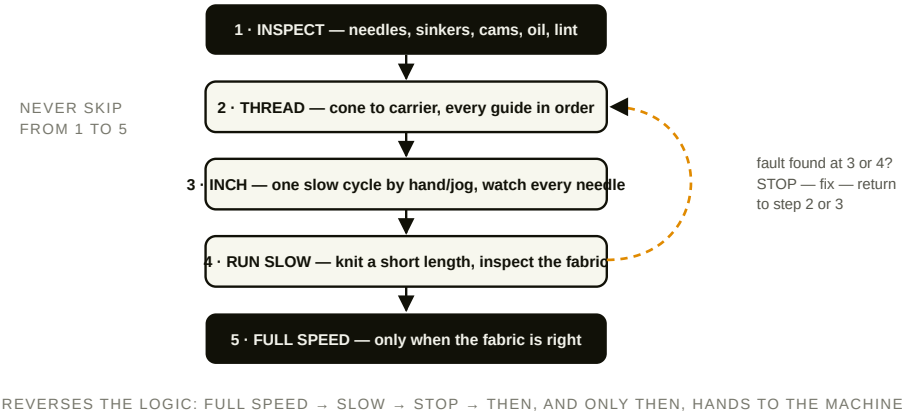


Figure 1 — The start-up sequence. Each step qualifies the machine for the next; a fault found while inching or running slow sends you back to fix it, never forward to full speed.

3. Threading

Thread one feeder completely and correctly rather than several feeders approximately. The yarn path (circular machine; adapt the order to the flat and warp machines from their guides) is:

cone → overhead guide eye → tension discs (yarn between the discs, around the post as the maker specifies) → stop-motion eye/feeler → feeder input guide → storage drum (check the coils build evenly and the reserve is present) → output guide eye → yarn carrier → needle hooks

1. Pull enough yarn from the cone to reach the needles without snatching; join to the previous yarn only with a small, flat weaver’s knot that will pass the knot detector — or thread fresh and knit the tail in, trimming it at the fabric.
2. Confirm the yarn passes *through* every element, in order, seated in the bottom of each guide: a yarn riding over a tension post has no tension control, and a yarn missing the stop-motion eye has

disconnected the machine’s protection.

3. At the carrier, set the yarn in the guide hole and check by inching that it is laid **into the open hooks** (Study 2): too high, it rides on the latches; too low or too far out, needles miss it.
4. Flat machine: thread via the mast tensioner to the carrier in use; park unused carriers clear. Warp machine: threading is beam work done at warping and tie-in; the operator’s duty is piecing a broken end back to *its own* guide and checking for crossed ends.

4. Starting, running and stopping

4.1 Starting

1. Complete the pre-start inspection (Section 2) and threading (Section 3).
2. **Inch** the machine by hand or jog control through at least one full knitting cycle at the feeder you threaded; watch clearing, feeding and knock-over at the needle line.
3. **Start slow** for a short length and inspect the fabric: every wale knitting, no ladders, holes or double loops, density even. Only when the fabric is right, **raise to full speed in steps**, listening as you go — a new rhythmic click or whine is a fault announcing itself; find it now, not at the doff.
4. Record the start: time, machine, yarn and count, structure, settings in use (Studies 1 and 3).

4.2 While running

- Patrol by eye and ear: fabric for ladders and holes, needle line for a needle riding high or a wale dropping, creel for cones running out or sloughing, feeder drums for reserves running low, frame and floor for oil where it should not be.
- Piece up a broken end promptly; after piecing, watch until the joint has knitted through.
- Never clear lint, free a needle or adjust anything at the needle line of a running machine — stop first (Section 7).

4.3 Stopping and doffing

1. **Normal stop:** use the stop control; let the machine come fully to rest before touching anything; leave the fabric under take-down tension, yarn threaded, so the next start is clean.
2. **Fault stop:** note which lamp is lit *before* resetting — it is the machine’s own diagnosis (Study 3). Correct the fault, inch once through the affected feeder, restart slow.
3. **Emergency stop:** for danger to a person or the machine only — press, do not hesitate, and report it; the machine is restarted only after the cause is known.
4. **Doffing:** at the set length or weight, stop, cut the fabric squarely above the roll (supporting its weight), remove and label the roll (machine, structure, yarn, date), start a fresh roll, record the doff.

5. Needle and timing checks

A machine is “in time” when every element arrives at the right place at the right instant. Re-timing is the mechanic’s work; the operator must *detect* mistiming and make the running adjustments. Checks to practise:

- **Needle-to-feeder timing (circular):** inching, confirm needles reach full clearing height just before the carrier presents yarn, and begin to descend just after it is laid. Yarn too early is laid on the latches (missed stitches, press-off); too late, the leading needles miss it (holes at one side of the feeder’s arc).
- **Sinker timing:** sinkers fully forward before the needles rise, releasing in step with knock-over. Fabric seen to *lift and fall* with the needles means late sinkers or a worn cam ring — report it in those words.
- **Stitch-cam equality:** all feeders at the same dial reading — and verify in the fabric, not on the dial alone: watch for barré (Study 3) and check stitch length by unravelling (Study 5).
- **Carrier position:** height and stand-off set so yarn enters the hooks centrally; re-check after any carrier change.
- **Flat machine:** brushes opening latches just before the carrier lays yarn; stitch cams of a bed pair equal unless the structure demands otherwise; racking returning the beds to exact opposition.
- **Warp machine:** swing and shog timed against needle-bar rise, the overlap completed while the hooks are clear of the sinkers. Any clashing of guides against needles: stop and report at once.

6. Fault finding during running

Fault finding is a method, not a guess: **read the fabric first** — its direction names the family before you touch the machine. **Along** the fabric (one wale, a ladder, a needle line): one needle or one yarn path. **Across** it (a band, a dropped course): a feeder, a cam, a whole cycle. **Everywhere**: the yarn, the settings or the take-down.

FAULT IN FABRIC	MOST LIKELY CAUSES	OPERATOR’S ACTION
Dropped stitch / ladder (runs along one wale)	Bent or broken needle hook; stiff or broken latch; sinker bent; yarn missed the hook at that needle	Stop; find the needle serving that wale; test its latch; replace the needle if damaged; mend the ladder back to the needle before restarting
Holes (scattered or in a line)	Yarn laid outside the hook (carrier out of position); yarn tension snatching the loop through; damaged needle; knot or slub bursting a loop	Check carrier position and feeding by inching; check input tension and cone state; inspect the needles of the affected wales

Press-off (fabric falls from needles)	Yarn break not detected; carrier empty; take-down failed so fabric rode up and off	Stop at once; re-hang or restart the fabric per laboratory practice; find why the stop motion did not act and report it
Barré (bands across the fabric, feeder repeat)	One feeder’s stitch cam or delivery different from the rest; mixed yarn lots between cones	Compare that feeder’s stitch-cam reading and measured delivery with the others; check cone tickets for a mixed lot
Needle lines (fine vertical streaks)	Individual needles tight or loose in their tricks; a needle of slightly different type fitted; lint in a trick	Identify the needles from the streak positions; clean tricks; replace non-matching needles with the correct type
Thick and thin places	Yarn irregularity; feeder reserve emptying (feeding gone negative); tension discs fouled	Check the drum reserve, clean the tension device; if the yarn is at fault, change the cone and report the lot
Oil stains	Over-oiling; oil flung from a cam box; a leaking oil line	Reduce oil feed to specification; clean; mark the stained fabric for the report
Warp machine: missing end / broken wale line	One warp end broken or its guide damaged; end not pieced to the correct guide	Stop; piece the end to its own guide (crossed ends make the next fault); check the guide for burrs

7. Safety — the rules that are never optional

- **Stop before hands.** No hand to the needle line, inside the cylinder, between rollers or into any drive while the machine can move. Inch with the jog control, never by pulling belts.
- **Isolate for maintenance.** Before any work beyond operating adjustments, switch off, isolate and — where the laboratory provides it — lock out, so no one else can start the machine while your hands are in it.
- **Clothing and hair.** No loose sleeves, ties, scarves, jewellery or uncovered long hair near rotating cylinders, shafts and carriage drives.
- **Guards stay on.** A guard removed for a check is replaced before restarting; an interlocked guard is never defeated.
- **Needles are sharps.** Handle loose needles by the stem, carry them in a container, and put every broken needle and fragment in the sharps tin — a needle fragment left in fabric or on the floor is an injury waiting for someone else.
- **Compressed air** only as the laboratory permits, never at skin, never blowing lint into another machine’s needle line. **Report, do not hide:** every accident, near-miss, defeated stop motion and

damaged guard is reported the same day — the operator who stops for a real doubt is right; the one who runs on is not.

8. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Give the start-up sequence in order, and state why full speed is the last step rather than the first.
2. During inching, yarn is seen lying across the latches instead of inside the hooks. Which two adjustments are wrong, and what fault would running on produce?
3. A ladder runs down one wale. Using the fault-finding method, say which family the fault belongs to and the first three things you check.
4. A machine stops with the yarn-break lamp lit but the yarn is intact. Following Study 3, give the correct operator response in order.
5. State three safety rules you would enforce before allowing a fellow student to change a needle on the circular machine.

Answer key

1. Inspect → thread completely → inch one full cycle watching the knitting point → run slow and inspect the fabric → raise to full speed in steps. Full speed is last because faults destroy needles and fabric in proportion to speed: a fault costing one course while inching costs hundreds of courses — and possibly a needle set — in the seconds a fast machine takes to stop.
2. The **carrier position** (yarn presented too high, meeting the latches before the hooks) and **needle-to-feeder timing** (yarn arriving before clearing is complete and the latches open). Running on gives missed stitches at that feeder — holes, or a press-off if needles miss together.
3. Along the fabric = **one needle or one yarn path**. Check: (1) that wale's needle — hook bent/broken, latch stiff/broken; (2) the sinker beside it; (3) the yarn's arrival at that needle. Correct, mend the ladder back, inch, restart slow.
4. Do not reset blindly: (1) read the lamp — a yarn fault at a specific feeder; (2) inspect that feeder's whole path — yarn out of the detector eye, bent feeler, fouled sensor, or tension so high the yarn rides out of the feeler; (3) correct the cause, re-seat the yarn through the detector, inch once, restart slow and watch the joint knit through.
5. Any three: machine isolated (switched off, not merely stopped) so no one else can start it; guards replaced before restart; needles handled by the stem, fragments into the sharps tin; no loose clothing or hair; replacement checked for type, gauge and free latch before inching.